

Semantic Context Effects in Visual Word Recognition, Sentence Processing, and Reading: Evidence for Semantic Strategies

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Earlier research studying the effects of semantic context on single words suggested that subjects may have two strategies for using a context (Becker, 1980). The present research finds that the semantic context strategies may be used in reading short sentences. Further, individual differences in context effects both in a word-level task and in a sentence-level task are related to individual differences in reading continuous text. These results are presented within the framework of the verification model (Becker, 1976, 1980), and the implications for two-process theory (Stanovich & West, 1979, 1981) are discussed.

The study of semantic context effects in visual word recognition has begun to address questions about how an appropriate semantic context facilitates the recognition of words and about how an inappropriate semantic context interferes with the recognition of words. The data show that subjects can respond to target words in a semantically related context both more quickly and more accurately than target words in a semantically neutral context. Also, subjects respond to words in an unrelated context more slowly and with more errors than words in a neutral context (Fischler & Bloom, 1979; Neely, 1976, 1977; Schuberth & Eimas, 1977; Stanovich & West, 1979, 1981). This refinement in the experimental analysis of semantic context effects has resulted in a comparable refinement in the theories used to explain con-

text effects. One such theory serves as the focus of this study.

To accommodate his data on facilitation and interference effects of a semantic context, Becker (1980) suggested that subjects may have two strategies for using the information provided by a semantic context. He proposed that subjects can either *predict* a small number of related words, or they can *expect* a larger number of possibly related words. In a lexical-decision (word vs. nonword) task, Becker found that facilitation and interference effects varied systematically as a function of the type of related context. For one type of related context materials, a *facilitation-dominant* pattern emerged. When stimulus lists used antonym pairs in the related context condition (e.g., context: HOT; target: COLD), a total context effect of 56 msec consisted of a 52-msec facilitation effect and only a 4-msec interference effect. In contrast, other materials produced an *interference-dominant* pattern. Here, a category name served as the context stimulus, and a member of the category served as a related target word (e.g., context: FURNITURE; target: COUCH). The category members were either common instances of the category or uncommon instances of the category. For this type of stimulus list, a total context effect of 70 msec consisted of a 9-msec facilitation effect and 61-msec interference effect. The dominance pattern was further shown to be independent of the specific items used in the related con-

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dition. A subset of the category items produced a facilitation-dominant pattern when mixed into a list including pairs of antonyms and pairs of strongly associated words.

An analysis of these stimulus materials as well as some materials used elsewhere (Fischler & Bloom, 1979; Stanovich & West, 1979) suggested that two modes of semantic processing could account for the difference in the pattern of facilitation and interference. First, the consistent use of strongly related word pairs induced subjects to respond to a context stimulus by considering only a limited set of words as candidates for a related target. Since only strongly related pairings are used, this use of the context ought to be successful on most, if not all trials. This mode of semantic processing was called the *prediction strategy*. Second, when the experiment includes a mixture of strongly, moderately, and weakly related context-target pairs, subjects are assumed to increase the number of words that are processed as related to the context. This should increase the likelihood that all words used in the related target condition will be included as candidates. This mode of semantic processing was called the *expectancy strategy*. The distinction between these two strategies, then, is the number of words primed by the context.

When the strategy assumption is incorporated into the verification model (Becker, 1976, 1979; Becker & Killion, 1977), that model can account for many of the details of the two patterns of data. In the original verification model, word recognition is the result of a successful match of a candidate word to the stimulus. The process of selecting a candidate and comparing it to the stimulus is viewed as a serial search operation using only one candidate at a time. Candidate words can be selected from one of two sets, one containing words semantically related to the context and one containing words consistent with a partial sensory analysis of the stimulus (e.g., a sensory feature analysis of line segments). The set of semantically related candidates is defined during the processing of the context stimulus, and thus, it can be complete and ready for use before the presentation of a target stimulus. When this set exists (i.e., when a semantic context is provided), the search process examines the

related candidate set as soon as the target stimulus is presented. The set generated from a sensory analysis of the target stimulus cannot be defined until after target stimulus presentation and after the analysis of the stimulus. This sensory candidate set is searched only after a related candidate set has been searched exhaustively. The assumed order of related set-sensory set search applies only when both sets exist. For words presented following a neutral context, no related set can be defined, and search begins with the sensory set.

Within this basic structure, the strategy assumption implies a difference in the size of the related set that is generated for a given context stimulus. Depending on the characteristics of the contexts used, the semantically related set can be large, or it can be small. For a small related set, a related target ought to be found after only a short search, since there are few competing words in the set, and thus a related target can be responded to quickly. Further, the processing of an unrelated target should be unaffected. Presumably, a small related set can be searched exhaustively while the target stimulus is undergoing sensory analysis, and when the sensory set becomes available, the search process can be ready to examine that set. Compared to the neutral condition, the processing of a small related set implies facilitation for related targets and no interference for unrelated targets—the facilitation-dominant pattern of the prediction strategy.

In contrast, a large related set produces longer recognition times for related targets and longer times for unrelated targets. Given a large related set, the time needed to search for and find a related target should be longer, on average, than the time needed to find a related target in a smaller set of words. Thus, related-target response times should show less facilitation for large related sets. For unrelated targets, a large related set may take a significant amount of time to search exhaustively. If this search time is sufficiently long, then the search of the sensory set will be postponed. That delay should be reflected in large interference effects for targets in unrelated contexts. Large related sets, then, produce only small facilitation effects coupled with large interference effects—the interfer-

ence-dominant pattern of the expectancy strategy.

With this description of word processing, the verification model can handle the data reported by Becker (1980). However, other experiments find a third pattern of facilitation and interference that the model cannot account for. Sometimes, the effect of a semantic context includes both a large facilitation component and a large interference component (Neely, 1977; Stanovich & West, 1979, 1981). Given the present structure of the verification model, this mixture of facilitation and interference cannot be explained.

There are two ways for the model to address large facilitation and interference effects within the same experiment. First, the size of a related set can be allowed to vary continuously from large to small. According to this view, the data reported by Becker (1980) would have been sampled from the extremes of the possible set sizes, whereas other data would have been sampled from intermediate set sizes. Second, the current assumption of just two set sizes can be retained, and intermediate facilitation-interference data can be attributed to averaging across some subjects who use the prediction strategy and other subjects who use the expectancy strategy. Thus, any experiment that shows large facilitation and large interference effects should include one group of subjects showing the facilitation-dominant pattern and another group of subjects showing the interference-dominant pattern. Averaged across groups, both facilitation and interference effects should be observed.

We can discriminate between the continuous size variation hypothesis and the discrete strategy hypothesis by examining their predictions about data for individual subjects. The continuous set size view holds that the data for each subject should match the group data. Each subject should show both facilitation and interference effects. Conversely, the discrete strategies view maintains that some of the subjects will produce facilitation-dominant data, whereas others will produce interference-dominant data. Only the group results will show both facilitation and interference effects. Of course, it could be that subjects vary their strategy from one trial of an experiment to the next. If so, the

discrete strategy position is indiscriminable from the continuous set size view in individual subject data, given present levels of experimental accuracy.

Given this analysis, we directed our efforts to investigating possible individual differences in semantic context effects. As a shortcut in the experiments that follow, we have adopted an empirical instead of a theoretical approach to identifying individual differences, but the theory still provides the framework within which to interpret any individual differences. The first step was to find a word-level semantic context task that produces both facilitation and interference effects in the averaged data. Pilot data indicated that the category materials used by Becker (1980) would produce this pattern when the subjects were told to use the category name cues to "predict" possible related target words. Here, both the facilitation and interference effects were significant; whereas without the instructions, the data were clearly interference-dominant. This finding alone provides some support for the proposed semantic strategies. Since instructions can have an effect on the pattern of facilitation and interference effects, it seems that subjects have some form of control over their use of context.

Next, we needed a measure of our subjects' performance on a related but independent task that could be used to identify individual differences. Individual differences in word-level semantic context effects should have their roots in subjects' normal reading behavior, and therefore, we selected a simple reading test as our second task. Since our primary measure of the word-level context effects is a reaction time measure, we chose reading time as the measure of reading performance. Also, we needed a distinction in reading materials that parallels the distinction between materials in the word-level task. As a first approximation, we selected two passages that varied in text difficulty. Some pilot subjects were tested with these passages to determine whether the difficulty factor produced a difference in reading rates and whether there were any apparent individual differences in the effect of difficulty. When instructed to "read or skim" the passages quickly for general information only, there

was a sizable effect of text difficulty averaged across all of the pilot subjects. Further, some subjects reduced their reading rate substantially on the more difficult passage compared to the easier passage, while other subjects read the difficult passage at nearly the same rate as the easier passage.

As we mentioned earlier, we found a pattern of significant facilitation and interference effects when subjects were instructed to use category-name cues to predict related category-member targets. If this pattern was the result of individual differences, then these individual differences could be related to individual differences in the reading task. Relying simply on relative speed of word recognition responses and reading, we would argue that all subjects can follow the instructions to "read or skim" quickly on the easy passage. Perhaps subjects are using the prediction strategy for this passage. Subjects who slow their reading rate for the difficult passage may be shifting from the prediction strategy on the easy passage to the expectancy strategy on the difficult passage. Other subjects, those whose reading rates do not vary, may use the same strategy—the prediction strategy—for both passages. It is this group of subjects who could be more skilled in using the prediction strategy and who might well be able to follow the prediction instructions in the category materials lexical-decision task. The other group of subjects are presumably less skilled at prediction and may need to use the expectancy strategy in the lexical-decision task.

The experiment just suggested clearly leaves behind many important questions. However, the experiment may begin to resolve the question of whether the suggested semantic strategies are qualitatively distinct or whether they differ on the quantitative measure of semantic set size. Moreover, the results may open the way for a generalization of word-level theories of semantic context effects. If we can demonstrate a relationship between semantic context effects in the word-level lexical-decision task and performance on a reading task, it would be among the first empirical links between research on word recognition and research on reading. Of course, it would be a tenuous, initial link, but it would still be an improvement over our

present *assumption* that word recognition has something to do with reading (see Mitchell, in press).

Experiment 1

Experiment 1 combines a reading test with the lexical-decision task used by Becker (1980). The reading task consisted of a practice passage, an easy test passage and a difficult test passage (see Miller, 1976). Performance on this task was used to assign subjects to one of two groups. One group included subjects whose reading rates were only slightly affected by the increase in text difficulty. The other group consisted of subjects whose reading rates dropped with increases in the difficulty of the text. All subjects who met the criteria for these reading groups then were tested in a lexical-decision task using related, neutral, and unrelated cue conditions with category materials. They were instructed to use the cue words to predict related target words.

Method

Subjects. A total of 57 University of Minnesota undergraduates participated in the reading test phase, and 24 of them participated in the lexical-decision task. Subjects were recruited from either the Introductory Psychology course or through advertisements in the campus newspaper. Those recruited from the course either received course credit or were paid for participating. The remainder were paid. Subjects were tested individually.

Reading test materials and procedure. Three passages and the associated "general idea" comprehension test questions were selected from Miller's (1976, pp. 131–134, 136–140, 142–146) materials. All three passages reflected various aspects of American society and were rated as familiar by Miller's subjects. One passage, rated at a 10th-grade level using the Flesch formula, served as a practice passage. Subjects were instructed to "read or skim" the approximately 800-word passage for a general understanding and not to worry about the detailed information. Subjects were then tested for comprehension of the passage by three multiple-choice questions. Each of the questions required only general knowledge of the content of the passage. Subjects were given feedback on their answers only for the practice passage.

The first test passage, rated as easy (seventh-grade level using the Flesch formula), contained 1,217 words, and the second test passage, rated as difficult (college sophomore level), contained 1,224 words. Five multiple-choice general comprehension questions followed each test passage. While taking the comprehension tests, subjects were not allowed to review the passage. For all passages, reading times were measured from the experimenter's signal to start until the subject indicated finishing.

Reading test results. The reading times were converted into reading rate scores using the number of words read per minute for each of the test passages. The effect of text difficulty was determined by subtracting the reading rate for the difficult passage from the reading rate for the easy passage. This result was then divided by the reading rate for the easy passage to yield a percentage-change score. This handling of the data eliminated overall differences in reading rate and focused on the relative effect of text difficulty. The percentage-change score was then used to assign subjects to reading groups. Twelve subjects who showed less than a 12.5% drop in reading rate across the two test passages were assigned to the "low change" group. Twelve subjects who showed a drop in reading rate of over 20% across the passages were assigned to a "high change" group. Thirty subjects did not meet these criteria and were not assigned. At this point in the research, we felt that our subject population should be selected from the tails of the reading test distribution to provide a clear distinction between the two subject groups (see, however, Experiment 3). Three additional subjects who met the criterion for the high change group served as pilot subjects in an unrelated experiment.

The main effect of text difficulty was reliable, $F(1, 22) = 56.28, p < .001$. Although the reading groups resulting from these assignments did not differ in overall reading rate, $F(1, 22) < 1$, this factor interacted with text difficulty, $F(1, 22) = 30.09, p < .001$, as expected. From the data shown in Table 1, the two reading groups clearly read the easy passage at the same rate. This outcome is consistent with the assumption that both groups used the prediction strategy for the easy passage and that the low change group also used the prediction strategy for the difficult passage.

Lexical-decision task procedure and materials. The procedures followed here are those used by Becker (1980) in his Experiment 2. Stimulus presentation, timing, and the collection of responses was controlled by an 8080-based microprocessor. Subjects were tested individually in a darkened, sound-attenuated room. A video monitor, used as the subject's display, was positioned at one end of a table, and the subject was seated at the opposite end with a response panel a comfortable distance away on the table. A single trial in the experiment consisted of a cue stimulus presented for 750 msec at a position slightly above the center of the video display. This was followed by a 300-msec blank display and then by a target stimulus presented slightly below the center of the display. The target stimulus remained visible until the subject responded. The target stimulus was an uppercase string of letters that either spelled an English word or did not spell a word. Subjects were instructed to read the target stimulus, decide whether it is a word or not, and press one button for words and another for nonwords. They were also told to make their response as quickly as possible while keeping errors to a minimum. The subject's response terminated the display of the target and started a 1.5-sec intertrial interval. On those trials when the subject made an error or failed to respond within 1.5 sec of the target onset, an error message was displayed for 1 sec, and the intertrial interval was timed from the end of the error message. In all, four blocks of 45 trials each were run with the first block serving as practice. Between blocks, there was at

Table 1
Mean Reading Rates (in WPM) for Each of the Groups in Experiment 1

Group	Text difficulty		Mean % change
	Easy	Difficult	
Low change			
WPM	447	428	4.3
% error	0	10	
High change			
WPM	465	340	27.1
% error	0	6.7	
Other subjects	420	352	18.6
	0	7.3	

Note. WPM = words per minute; % error = error rate on comprehension test. The data for "Other subjects" are the results for the 30 subjects who did not meet the high change or low change criteria.

least a 15-sec rest period before the subject was allowed to begin the next block of trials.

The cue stimulus on a given trial could be either a word or a string of five Xs serving as a neutral cue. Subjects were told that the neutral cue indicated they were to get ready to read the target and make their response. A word cue would name a category of things (e.g., DOGS). Further instructions emphasized that if a word target followed the word cue, the target was likely to be a member of the category named by the cue. The subjects were told to use the category-name cue to generate a list of category members that might be presented as the target. As an example, given a category cue of DOGS, they might generate the list COLLIE, BOXER, PEKINESE, POODLE, and so on. They were told to work quickly because there was only about 1 sec between the onset of the cue and the onset of the target. Subjects also were warned that a word target following a word cue would not always be related and that the target would not always be a word.

The stimulus materials were those used by Becker (1980) in his Experiment 2. For the test materials, the entire set was divided into a set of critical items and a set of filler items. The critical materials consisted of 18 category names and three members of each category selected from the Battig and Montague (1969) norms. The category names were those that can be represented by a single word (e.g., BIRDS, FURNITURE, RELATIVES). The category members were selected to be highly typical of the category (given as a response to the category name by an average of 298 of the normative subjects), moderately typical (responses from 191 of the subjects), and low on typicality (responses from 57 of the subjects). Also, across the levels of category typicality, the mean frequency of occurrence is nearly equal (51.2, 54.8, and 48.6 Standard Frequency Index [Carroll, Davies, & Richman, 1971] for the high-, moderate-, and low-typical item sets, respectively).

The assignment of critical stimulus materials to the various cuing conditions assured that each critical target word appeared equally often in each cue condition for each of the reading groups. For a given subject, one third

of the target words at each of the levels of category typicality were used in the related cue condition (e.g., FURNITURE-CHAIR). Another third of the target words were cued with the neutral cue (e.g., XXXXX-AUNT), and the final third were cued with an unrelated category name (e.g., FURNITURE-SHIRT). For other subjects, the assignment of target words to cue conditions was varied. A second subject saw the target word, CHAIR, following the neutral cue, and a third subject saw CHAIR following an unrelated category name. Thus, within a reading group, there were three subgroups of subjects that differed in the target words used in the three cue conditions. Across the reading groups, the same assignment of materials to cue conditions was used.

The filler word target materials were selected from an additional nine categories in the Battig and Montague (1969) norms. For a related cue filler condition, the category names, one high-typical category member and one low-typical category member were used. In addition, two more members of each of the nine filler categories were selected as neutral cue fillers. These word targets were presented following the neutral cue. For the word targets, overall, there were 18 critical related cue trials, 18 critical neutral cue trial, 18 critical unrelated cue trials, 18 filler related cue trials, and 18 filler neutral cue trials.

In addition to the word targets, 45 nonword targets were developed by selecting 45 words from the category norms and changing one or two letters in the words. For example, the word COUSIN was misspelled as COUZIN. If possible, the misspelling preserved the pronunciation of the word. Twenty-seven of the nonwords were presented following the 27 category-name cues used in the word target conditions. The remaining 18 nonwords were presented following the neutral cue.

Results

The initial analysis of the lexical-decision data considered only correct responses to the critical target words. Those trials on which the subjects failed to respond within 1.5 sec were excluded. These trials accounted for less than .1% of the data.

Table 2 shows that the results for the low

change reading group were facilitation-dominant for the lexical-decision task. Reaction times in the related cue condition were 43 msec faster than for the neutral condition and times for the unrelated condition were only 13 msec slower than the neutral condition. Subjects in the high change group produced the opposite pattern with interference dominating. For this group, facilitation was only 23 msec, and interference was 78 msec. This interaction of Reading Group $\times$ Cue Condition was reliable, $F(2, 22) = 6.98$, $p < .005$. The main effect for cue condition was also significant, $F(2, 22) = 39.33$, $p < .001$. Averaged across reading groups, the related cue condition was 33 msec faster than the neutral cue condition, and the unrelated cue condition was 46 msec slower than the neutral condition. The reading groups did not show a main effect, $F(1, 22) = 1.54$, $p > .05$.

The error data from the lexical-decision task showed only a significant effect of cue condition, $F(2, 44) = 9.47$, $p < .001$. The error rates were .4%, 1.6%, and 5.1% for the related, neutral, and unrelated cue conditions, respectively. This pattern of errors did not differ substantially for the two groups of subjects.

Discussion

The results of Experiment 1 supported the verification model description of the prediction strategy. In previous research (Becker, 1980), the same stimulus materials led subjects to select an expectancy strategy, that is, to produce interference-dominant data. In

Table 2
Mean Reaction Times (in msec) and Percentage of Errors for the Lexical-Decision Task of Experiment 1

Group	Cue condition			Facilitation (N - R)	Interference (U - N)
	R	N	U		
Low change					
RT	639	682	695	44	13
% error	.4	.9	4.6		
High change					
RT	665	688	766	23	78
% error	.4	2.3	5.6		

Note. R = Related; N = Neutral; U = Unrelated; RT = reaction time. Data are for the critical target items only.

the current experiment, the instructions to use the category-name cues to predict possible related targets created a shift in the pattern from interference dominance to a pattern showing both facilitation and interference effects.

A second result of Experiment 1 was that the average data from the lexical-decision task appeared to be the result of two different groups of subjects. One reading group contributed a substantial facilitation effect and only a small interference effect. The other group contributed a nominal facilitation effect and a sizable interference effect. This result is consistent with the view that experiments showing both facilitation and interference effects may reflect an averaging across individual differences in semantic strategy selection. This, in turn, supports the position that the semantic strategies are qualitatively distinct modes of semantic processing. Other data also support this conclusion. Becker (Note 1) reported an experiment using his antonym and category materials (Becker, 1980) and adding a visual intensity factor similar to that used by Becker and Killi6n (1977). For the facilitation-dominant antonym materials, dim target stimuli showed more facilitation than did bright target stimuli. In contrast, for the interference-dominant category materials, the effect of context condition was additive with the effect of intensity. These data provide some converging evidence supporting a clear distinction between the two strategies.

A third finding was that a subject's performance on the reading task and on the lexical-decision task were related. Subjects who read both the easy and difficult passages at a fast rate showed facilitation dominance in the lexical-decision task. Conversely, subjects who slowed their reading rate for the difficult passage seemed unable to follow the prediction instructions in the word-level task. These data should be taken as encouragement for additional efforts aimed at relating performance in word-level tasks to reading performance. Obviously, given the crude analyses of the word materials and the reading test materials, one cannot confidently describe the form of such a relationship.

A qualification must be stated for this conclusion. The error data for the lexical-deci-

sion task showed an interference-dominant pattern regardless of reading group. A similar result was reported by Becker (1980). There, error rates for category materials showed interference dominance for facilitation-dominant reaction times as well as for interference-dominant reaction times. This pattern of errors may be consistently produced for category materials regardless of the subjects' strategy.

One implication of the results of Experiment 1 is that the theoretical structures developed to describe semantic context effects in word-level experiments may generalize to units of text larger than single words. We might assume that when reading rapidly, subjects use something like the prediction strategy to take advantage of context. One segment of text may be used to predict the next segment of text. In contrast, when a subject reduces the rate of reading, we might assume that the expectancy strategy is being used. Here, the amount of time needed to recognize a text segment in context should be increased, yielding a decrease in the rate of reading. Of course, the size and characteristics of a text segment remain to be determined, as do the factors that control strategy selection.

Is it reasonable to generalize word-level theories to reading? The remainder of this article addresses this question. Specifically, Experiments 2-4 focus on three points. First, can the facilitation-dominant and interference-dominant context effects be obtained in tasks that require subjects to respond to more than a single word? Second, are the individual differences in semantic context effects robust enough to be detected in tasks other than the lexical-decision task? Third, are the descriptions of the prediction strategy and the expectancy strategy appropriate to units of text larger than single words?

Experiment 2

The purpose of Experiment 2 was to determine whether facilitation-dominant and interference-dominant context effects could be obtained when subjects were reading and responding to more than a single word. The task required subjects to decide whether a short sentence stated a true fact or not. This

Table 3
Summary of the Materials and Design for Experiment 2

Cue condition	Example		Probability of condition
Positive response trials			
Related	SNOW FALLS IN WINTER	SUMMER IS HOT	.27
Neutral	XXXXX	SUMMER IS HOT	.27
Unrelated	MILK COMES IN BOTTLES	SUMMER IS HOT	.13
Negative response trials			
Related	BANANAS ARE YELLOW	A PEACH IS A VEGETABLE	.13
Neutral	XXXXX	DRILLS HAVE CHAIRS	.13
Unrelated	A RAZOR CAN CUT	AN ARM IS A TOE	.07

Note. The last column gives the overall probability of each type of cue condition.

task was combined with the types of context conditions used earlier. If the results for this task parallel previous work with the lexical-decision task (Becker, 1980), we expect the strongly related list to show facilitation-dominant data and the mixed list to yield interference-dominant data.

Method

Subjects. Forty-eight University of Minnesota undergraduates were recruited from the same sources used for Experiment 1. None had participated in similar experiments. The first 24 subjects to appear for the experiment were tested using the strong related list of stimulus materials, and the second 24 subjects saw the mixed strength list.

Procedure. A Data General computer controlled the timing of events, the subject's display, and the recording of responses. Subjects in this experiment were tested either individually or in pairs in a darkened, sound-attenuated room. Each subject station included a video monitor located at one end of a table and a response panel positioned a comfortable distance in front of the subject at the opposite end of the table. The two subject stations were separated by a freestanding divider.

The subjects were instructed that each trial consisted of a warning cue followed by a target sentence. The cue, presented toward the top of the display screen, remained on for 1,750 msec. The screen was blanked for 300 msec, and then the target sentence was presented toward the bottom of the screen. All cue and target stimuli were composed of uppercase letters. Subjects were instructed to read the target sentence and to decide whether it was true or false. If it was true, they were to press the right-hand button. If it was false or did not make sense, they were to press the left-hand button. Subjects were also told to respond quickly while making as few errors as possible. The response terminated the display of the target stimulus. Reaction time was measured to the nearest msec from the onset of the target display to the response. Subjects were allowed up to 3 sec to respond to the

target. If the subject made an error or if the 3-sec limit was exceeded, an error message was presented for 1 sec. A 1,750-msec intertrial interval was measured from the response of the slower subject or from the offset of the error message. Subjects were also informed about the possible relationships between cues and targets. Examples of each type of cue-target pair were given as part of the instructions.

Each subject saw a block of 23 practice trials followed by three blocks of 30 test trials. Between each block there was a rest period of at least 15 sec. The entire session lasted about 25–30 min.

Stimulus materials. The stimulus materials included a set of critical items and several types of filler items. All the analyses are based on the critical items. Thirty-six semantically related sentence pairs were chosen as critical cue-target items. All sentences were simple, three- to six-word sentences. The complete list of these materials is given in the Appendix, and examples are shown in Table 3. In developing these materials, we attempted to select sentence pairs in which the target sentence was obviously related to the cue sentence but was not entirely predictable given the cue. Most of the critical target sentences were selected from a set of materials developed for pilot work. The earlier materials included two cue sentences, one of which had been rated as strongly related to the target and one of which was rated as weakly related to the target by a group of six judges. In developing the cue sentences for use in the present experiment, we attempted to write sentences that we judged to be intermediate between the strong and weak cues in the pilot materials.¹

¹ In selecting the materials for this experiment, we relied on our own intuitions about how the materials should be structured and on the results of extensive pilot work. We make no claim to have identified any of the potentially important stimulus factors. We claim only that the materials we have used will consistently produce the data that we report. This is, of course, a less than ideal approach, but to explore both the characteristics of the procedure we have selected and the intricacies of the materials is beyond the scope of a single study.

Table 4
Mean Reaction Times (in msec) and Percentage of Errors for the Sentence Task of Experiment 2

List	Cue condition			Facilitation (N - R)	Interference (U - N)
	R	N	U		
Strong					
RT	1,042	1,129	1,129	87	0
% error	1.3	3.1	3.8		
Mixed					
RT	1,034	1,082	1,139	48	57
% error	2.0	2.4	3.1		

Note. R = Related; N = Neutral; U = Unrelated; RT = reaction time. Data are for the critical target items only.

The critical target items were counterbalanced across the three cuing conditions. Specifically, the critical targets were divided into three sets of 12 items each, and across subjects, each of the item sets was used equally often in each of the three cuing conditions. For example, one subject saw the target A FOREST HAS TREES preceded by its related cue sentence. For another subject, the neutral cue preceded the sentence, and for a third subject, an unrelated sentence like SAILORS SERVE IN THE NAVY cued the target A FOREST HAS TREES.

Along with the 36 critical items, there were 12 related cue and 12 neutral cue filler items requiring a positive response and 30 negative-response items. The positive-response filler items served two purposes. First, they were used to increase the probability of related and neutral condition trials to match the probabilities used by Becker (1980). Second, the related fillers were used to vary the semantic composition of the overall stimulus list. Two lists of stimulus materials were constructed differing only in some of the items used in the related filler condition. For one list, all these filler items were *strongly* related pairs of sentences. In the other list, these filler items were a *mix* of strongly and weakly related sentence pairs. All other items were identical across the two lists. This includes the critical stimulus pairs on which the results are based. The negative-response items included sentences that ranged from obviously nonsensical (e.g., DRILLS HAVE CHAIRS) to those that are close to correct (e.g., A PEACH IS A VEGETABLE). Twelve of these negative targets were preceded by a related cue sentence, 12 by the neutral cue, and 6 by an unrelated sentence cue. Examples of each of the conditions are also included in Table 3.

Results and Discussion

Since we had explicit predictions, the results were analyzed with planned comparisons. Our model predicted that the strong list of materials would induce subjects to use the prediction strategy, yielding a facilitation-dominant pattern of results. Indeed, for this list, there was a significant 87-msec facilitation effect for correct responses, $t(23) = 2.738$, $p < .05$ (see Table 4). The nonsignificant interference effect was 0 msec, $t = 0$.

The data for the mixed stimulus list also are presented in Table 4. For these data, we expected subjects to show an interference-dominant pattern. A comparison of the related and neutral conditions showed a significant 48-msec facilitation effect, $t(23) = 2.42$, $p < .05$. The comparison of the unrelated and neutral conditions showed a significant 57-msec interference effect, $t(23) = 2.64$, $p < .05$. Thus, although we predicted an interference-dominant effect, the results suggested both facilitation and interference. Analyses of the error data yielded no significant effects for either list.

Following our earlier arguments, this type of result implies that individual differences affected the pattern of facilitation and interference. Some subjects in the mixed list condition may have selected the prediction strategy, whereas others opted for the expectancy strategy. To assess this implication, we conducted a post hoc analysis of the mixed condition data for individual differences in facilitation and interference effects. For each subject, we computed a "dominance score" by subtracting the facilitation effect from the interference effect. Within each of the stimulus materials conditions, subjects were divided into two groups using a median split of the dominance scores. The group with the lowest dominance scores showed a 92-msec facilitation effect and a -6-msec interference effect. The other group showed only 5-msec facilitation and 119-msec interference. (Because of the counterbalancing of stimulus materials across cue conditions, it was impossible to determine whether the overall distribution was bimodal.) Across the two groups mean reaction time in the neutral

context condition was about the same. This post hoc analysis suggests that there may have been individual differences in strategy selection. Granted, the method of identifying the individual differences was not independent of those differences, but the results were consistent with the analysis of Experiment 1.

Experiment 3

To further explore the possibility of individual differences, Experiment 3 combined the reading test of Experiment 1 with the mixed list condition of Experiment 2. The reading test should allow us again to identify a low change group of readers and a high change group of readers. For the mixed-list sentence verification task, the low change group ought to produce a facilitation-dominant pattern. The high change group should yield an interference-dominant pattern for the mixed-list materials. This outcome would provide added support for several of the points made here. The replication of the reading task data would validate the type of reading measure used earlier. The independent identification of interference-dominant subjects in the sentence task would provide the evidence necessary to generalize this pattern and the associated expectancy strategy to the level of sentences. Finally, the correspondence of reading and sentence task performance would lend added credibility to the conclusion that these kinds of context effects are directly related to reading performance.

Method

Subjects. Thirty-two subjects were recruited from the sources used in Experiments 1 and 2. None had participated in similar experiments.

Procedure. Each subject participated in a reading task and a sentence processing task, in that order. The materials and procedures for the sentence task were the same as those used in Experiment 2. The reading task was the same as that used in Experiment 1.

Assignment to reading groups. The reading time data from the reading task were converted to reading rates in words per minute for each of the test passages. Each subject's reading rate for the difficult passage was subtracted from the rate for the Easy passage. This difference was then divided by the rate for the Easy passage to yield a percentage change score. Within each of the three sentence-task materials groups, subjects were rank-ordered using the percentage change score. The five subjects in each of the three materials groups with the lowest scores

were assigned to a low change group, and the five subjects in each materials group with the highest scores were assigned to a high change group. Thus, we have one group of 15 subjects who were relatively unaffected by the text difficulty factor and another group of 15 subjects whose reading rates decreased on the difficult passage. Two subjects were eliminated from the further analyses to maintain the balancing of the sentence task materials.

The data from the reading task are presented in Table 5. The low change group averaged only a 9% decrease across the two passages; whereas the high change group averaged about a 23% decrease in reading rate. An analysis of variance on these data showed no main effect of group, $F(1, 28) < 1$, a significant effect of text difficulty, $F(1, 28) = 121.68$, $p < .001$, and an interaction of Group $\times$ Difficulty, $F(1, 28) = 27.10$, $p < .001$. The latter result was expected because of the method of assigning subjects to groups.

Results and Discussion

The sentence task data were analyzed using reading group and cue condition as the factors. For these data, we expected subjects in the low change group to produce facilitation-dominant results and subjects in the high change group to show interference dominance.

The data from the sentence task are shown in Table 6 as a function of reading group and sentence cue condition. For the low change group, the planned comparisons showed a significant facilitation effect of 87 msec, $t(14) = 3.99$, $p < .01$ (all significance levels are for one-tailed tests), and no interference effect for this group (-19 msec), $t(14) = 1.02$. For the high change group, there was no facilitation effect (19 msec; $t < 1$) and a significant interference effect of 70 msec, $t(14) = 3.36$, $p < .01$. The low change group

Table 5
Mean Reading Rates (in WPM) For the Two Groups in Experiment 3

Group	Text difficulty		Mean rate
	Easy	Difficult	
Low change			
WPM	335	306	321
% error	0	12.0	
High change			
WPM	357	275	316
% error	1.3	4.0	

Note. WPM = words per minute; % error = error rate on comprehension test.

Table 6
Mean Reaction Time (RT) and Percentage of Errors on the Sentence Task for the Two Reading Groups in Experiment 3

Group	Cue condition			Facilitation (N - R)	Interference (U - N)
	R	N	U		
Low change					
RT	1,052	1,139	1,120	87	-19
% error	1.1	2.8	3.3		
High change					
RT	1,110	1,129	1,199	19	70
% error	.5	2.7	1.6		

Note. R = Related; N = Neutral; U = unrelated.

yielded significantly more facilitation than the high change group (68 msec), $t(28) = 1.93$, $p < .05$, whereas the high change group produced significantly more interference than did the low change group (89 msec), $t(28) = 3.17$, $p < .01$. There were no significant effects in the error data for error rates of 1.1, 2.8, and 3.3% for the low change group and .5, 2.7, and 1.6% for the high change group (the error rates are for the related, neutral, and unrelated cue conditions, respectively).

The outcome of Experiment 3 supported the semantic strategy interpretation and the generalization of the strategies to normal reading. Although the measures of the subjects' reading performance are not refined, the use of a percent change score seems appropriate to identifying individual differences in strategy usage. In two instances, this measure of reading performance has been related to independent measures of the semantic strategies.²

Experiment 4

In interpreting the results of Experiment 3 and earlier experiments, we assumed that the correspondence of the data to an explanation based on semantic strategies constituted support for that view. Experiment 4 attempted a more direct test. If the semantic strategies are used in the sentence task, then we ought to be able to influence a subject's strategy selection through instructions. Experiment 1 indicated that instructing subjects to use a cue predictively affected facilitation and interference for single words. If the strategies apply to the reading of short sentences

as well as to the reading of single words, then instructing subjects to "predict" in the sentence task ought to produce results comparable to those for single words. In Experiment 4, we tested this implication using the mixed list materials from Experiment 2 and instructions to use the cue sentence to predict possible related targets. In the earlier uses of the mixed list materials, both facilitation and interference effects were obtained. Should the prediction instructions prove ineffective, the same results should obtain here. However, if the instructions induce all subjects to select the prediction strategy for these materials, then we should find a clear facilitation-dominant pattern.

Method

Subjects. Fifteen friends and acquaintances of the second author served as subjects. They ranged in age from 20 to 51 with most between the ages of 23 and 28. Eleven of the subjects were employees of the New Jersey Marine Sciences Consortium, Sandy Hook, New Jersey. All subjects had at least one year of college-level education with only one enrolled as a student at the time of testing. None of the subjects had degrees above the master's level.

Procedure. Presentation of stimuli, timing of events and the recording of data were controlled by a TRS-80

² We performed another analysis of our data to identify a group of fast readers and a group of slow readers among our subjects, using average reading rates across the two test passages. For these two groups there was no difference in the pattern of facilitation and interference in the sentence task ($F < 1$ for the interaction of Group $\times$ Cue Condition). Each of these groups showed a mixture of facilitation and interference effects and differed only in the speed of their responding, with the fast reader group about 50 msec faster in all conditions.

Table 7
Mean Reaction Time (RT) and Percentage of Errors for Experiment 4

Group	Cue condition			Facilitation (N - R)	Interference (U - N)
	R	N	U		
Prediction instructed					
RT	1,095	1,185	1,166	90	-19
% errors	1.1	2.1	1.6		

Note. R = Related; N = Neutral; U = Unrelated.

Model III microcomputer. Stimuli were presented in uppercase letters on the standard display of the computer, and responses were made using the keyboard. True responses were assigned to the / (slash) key in the lower right-hand corner of the keyboard, and false responses were assigned to the z key in the lower left-hand corner.

The instructions of Experiments 2 and 3 were modified to emphasize that cue sentences were to be used to predict possible related targets. Specifically, subjects were told to "use the information in the cue sentence to try to predict the content of possible related sentences." They were also asked to continue this type of prediction even though it might seem counterproductive at times (e.g., given a long run of unrelated or weakly related trials).

In all other facets, the procedures are identical to those used in Experiments 2 and 3. All the timing intervals used earlier were used here, and the mixed-list stimulus materials defined for Experiment 2 were used in exactly the same way as in the earlier studies.

Results

As in the earlier experiments, reaction times for correct responses to the critical target sentences and errors on the critical sentences were analyzed using planned comparisons to assess the facilitation and interference effects. The data, presented in Table 7, showed a significant facilitation effect of 90 msec, $t(14) = 4.97, p < .001$. The interference effect was not significant (-19 msec), $t(14) = -1.27, p > .05$. In these data, unrelated target sentences were responded to slightly faster than were targets in the neutral condition. Overall, the error rates were about 2% or less, and there were no significant effects in the error data for the critical target sentences.

In questioning subjects after their sessions, all but one stated that they had tried to follow the prediction instructions throughout the session. Almost all subjects spontaneously

reported that they thought their responses were faster to targets following neutral cues than to targets following sentence cues.

Discussion

The results of Experiment 4 conformed to our expectations about the effects of the prediction instructions. For the mixed list materials, the prediction instructions apparently induced a shift from a pattern showing both facilitation and interference to a pattern of facilitation dominance. In fact, the data are similar to those from the strong materials list of Experiment 2, as might be expected. In both experiments, we assumed that subjects used the prediction strategy, and since the critical target materials were the same, the results ought to be much the same.

Another possible interpretation of Experiment 4 and of some of the other findings needs to be considered. The instructions or changes in stimulus list composition might induce response biases. For instance, instructions that emphasize predicting related target sentences might produce a bias toward responding positively when the cue and target sentences were related and responding negatively when they were unrelated. This form of bias predicts that the error rate for unrelated target sentences should be quite high, a prediction that is inconsistent with the error data shown in Table 7. The error rate for unrelated target sentences was somewhat less than the error rate for sentences in the neutral condition and not much greater than that for related targets.

Another form of bias may have come from the perceived validity of the sentence cues either because of instructions or because of

highly predictive cue sentences. These factors could have induced a bias toward responding positively following a sentence cue and negatively following a neutral cue. Such a bias would slow correct responses to true sentences following the neutral cue and would shift the data toward a pattern of facilitation dominance. This form of bias predicts that negative responses should be faster and errors fewer for false sentences following the neutral cue compared to false sentences following a sentence cue. The false response data, shown in Table 8, do not support this prediction. Subjects responded to false sentences following sentence cues both faster and often with fewer errors compared to neutrally cued false sentences. Moreover, the pattern does not appear to change as a function of the pattern obtained for the true-response materials. In making comparisons across the context conditions, bear in mind that particular false target sentences are confounded with the cue condition.

General Discussion

The major implication of the present findings is that the strategy distinction developed for word-level context effects appears to accommodate units of text larger than single words. We have shown that the results of Becker (1980) for single words can be obtained when subjects are responding to short sentences. We also have shown that an individual subject's performance in a word recognition task and in a sentence task is related to one measure of that subject's reading performance. Finally, for both single words and short sentences, we have shown that subjects can be instructed to use one strategy—the prediction strategy—when they might otherwise choose the expectancy strategy.

Obviously, we need to specify how to generalize the verification model and its strategies. Given our approach in the current study, though, we have neither the type of materials analyses nor the associated data to state the generalization formally. The current study achieves another goal, however. It now seems clear that the strategies described by Becker (1980), or some minor variant of them, can be used to describe how subjects process short sentences in context and how

Table 8
Mean Reaction Time (RT) and Percentage of Errors for False Response Trials

Group	Cue condition		
	R	N	U
Experiment 2			
Strong list			
RT	1,290	1,311	1,209
% error	9.0	8.3	5.5
Mixed list			
RT	1,239	1,273	1,190
% error	10.0	9.7	1.3
Experiment 3			
Low change			
RT	1,278	1,309	1,230
% error	6.1	11.1	3.3
High change			
RT	1,376	1,398	1,344
% error	9.4	3.3	6.7
Experiment 4			
RT	1,295	1,345	1,231
% error	9.4	7.2	2.2

Note. R = related; N = neutral; U = unrelated.

subjects might use context in reading short text passages.

Although the results of the current study supported the verification model, several issues about word recognition remain unresolved. To highlight these issues, we now turn to consider an alternative model, the two-process theory described by Posner and Snyder (1975) and others (Neely, 1976, 1977; Stanovich & West, 1979, 1981). As we shall see, two-process theory encounters considerable difficulties in accommodating the data reported here. However, the verification model encounters difficulties of its own in handling the data that best support two-process theory.

In two-process theory, an automatic, spreading-activation process is assumed to produce facilitation for words related to a context without producing interference for words unrelated to the context. Interference effects are attributed to a subject's use of a limited-capacity attentional process. When a subject directs attention to context information, subsequent stimuli that do not fit the

context suffer the cost of switching attention from the context to the stimulus before a response can be made. More specific to present purposes, Posner (1982) suggested that the facilitation-dominant pattern described by Becker (1980) implies that subjects choose to process the context automatically. The large facilitation and small interference effects here follow directly from the description of automatically processing the context. Conversely, the interference-dominant pattern indicates that subjects have chosen to attend to the information provided by the context. Here, the large interference effect indicates the use of attention, and the small facilitation effect can be attributed to an averaging across trials on which subjects focus attention on the correct aspect of the context and trials on which they focus incorrectly.

In spirit, Posner's suggestion is comparable to the semantic strategy assumption made by Becker to account for the same data. However, the two positions differ in their predictions for experiments like Experiments 1 and 4 of the current study. As stated above, Becker's verification model suggests that instructing subjects to focus on the context and use it predictively should increase facilitation while decreasing interference. Posner's view, however, predicts that the instructions ought to increase the interference effect as more subjects choose to attend to the context, or at least, that the instructions should not decrease the observed interference. The data from Experiment 1 and more clearly from Experiment 4 are consistent with the verification model prediction and disconfirm the two-process theory prediction.

Since the present data do not support a prediction derived from a basic assumption of two-process theory, namely that directing attention to a context results in interference, we need to consider some of the other reasons for that assumption. Two-process theory received substantial support from a study reported by Neely (1977). That study marked the first time that a large interference effect was obtained along with facilitation in an experiment examining semantic context effects. It was also the first time that subjects were directly and repeatedly instructed to pay attention to the context. When these results were reported, it seemed reasonable to con-

clude that the instructions produced the interference effect. However, Neely's study also marked the first use of category materials, materials that Becker (1980) has shown to be interference-dominant, given no special instructions. Based on the more recent evidence, then, the conclusion could well be that the category materials produced the interference effect, whereas the instructions led to the facilitation that Neely observed. Thus, Neely's basic findings can be interpreted in either framework.

There are other facets of Neely's data that supported two-process theory. First, as the interval between the onset of the context and the onset of the target stimulus increased, both facilitation and interference increased. Two-process theory assumes that it takes time to direct attention to the context, and therefore, attentional effects are more likely to occur at longer intervals than at shorter intervals. Second, Neely instructed his subjects to treat one category name as a cue that a member of a different category was likely to occur as the target. At short cue-target intervals, members of the category named by the category-shift cue were facilitated, whereas members of the shifted-to category were not. This finding was interpreted to mean that members of the named category were automatically activated by the occurrence of the category name, but the shifted-to category members could not be automatically activated. As the cue-target interval increased, members of the shifted-to category showed facilitation, whereas members of the named category showed interference. As time allowed, subjects presumably directed their attention to the shifted-to category and thereby inhibited members of the named category.

Neither of these points can be accommodated by the verification model. That model has been used to describe the "steady-state" operation of word recognition processes when subjects are under only nominal time pressure, and it has not been applied to the results of variations in timing parameters. Also, the model has not been applied to Neely's findings on the redirection of attention from one category to another. Obviously, these are shortcomings in the presentation of the model. As we explore vari-

ations in semantic strategy usage across cue-target intervals, however, we may begin to address at least the effects of timing parameters.

As an example of what can be expected from this effort, West and Stanovich (1978) have argued that two-process theory can describe the effects of context for both adult subjects and children. Their data showed significant facilitation and interference effects for children but only a facilitation effect for adults. The West and Stanovich account of these data focuses on the differences in response times for the two groups of subjects. According to two-process theory, the slower responses of the children allowed enough time for attentional interference effects to emerge, whereas the adults in the study responded quickly enough that they did not have time to direct their attention to the context. Although these data are accurately described by two-process theory, Becker (1982) has shown that children can show a facilitation effect without any interference and that they can show an interference effect without any facilitation. In other words, children show the same facilitation-dominant and interference-dominant patterns as do adults. Extending the conclusions of the current work, the mixed facilitation and interference results reported by West and Stanovich may reflect individual differences in strategy selection by the children in that study. Thus, the time differences that are important in a two-process theory account of the data may be replaced by strategy selection differences in a verification model account.

Still other difficulties exist for each model. Stanovich and West (1981) reported two experiments in which a context factor interacted with a word difficulty/frequency factor. As they argued, two-process theory can account for the interaction if the extra time needed to respond to the more difficult words allowed a subject's attention to operate. This assumption predicts the interaction and an interference effect for the difficult words as well. The verification model also predicts the interaction of context and word difficulty/frequency (cf. Becker, 1979; Sharkey & Mitchell, Note 2), but this model does not predict an increase in interference for difficult words. In their Experiment 2 and Ex-

periment 3 (0-msec delay between cue offset and target onset), Stanovich and West reported the interaction of context and word difficulty, but in both cases there was no difference in interference effects across levels of difficulty. This outcome is not as predicted by two-process theory; however, it is consistent with the verification model. In another condition of Experiment 3 (800-msec delay), the interaction of context and word difficulty was also found, but here, the more difficult words resulted in more interference than the less difficult words. This finding was as predicted by two-process theory, but the verification model can accommodate it only by assuming some strategy shifting during the delay.

The kinds of problems noted for the verification model and for two-process theory extend to other models as well. For example, Morton's (1969) logogen model can address some of the data reported in recent studies of word recognition (Schuberth & Eimas, 1977), but it fails in a number of cases (Becker, 1976, 1979, 1980; Becker & Killion, 1977). McClelland's (1979) cascade model handles many of the results from word superiority experiments, but it can only begin to address other data (Schvaneveldt & McDonald, 1981). It is becoming apparent that there is some truth in each of these models and that none can be rejected easily. What seems needed is a synthesis of opinions about the word recognition issues raised here and other issues about how words are processed. This complex task is made even more difficult by the results of the present experiments. Here, we have supported the generalization of aspects of word-level explanations to the processing of text units larger than single words. Thus, in addition to resolving the word-level issues, we also must consider whether—and specifically how—the resolutions are compatible with the processing of larger text units.

Reference Notes

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Appendix: Sentence Task Materials

Cue	Target	Cue	Target
Critical materials		Strong related fillers	
RACQUETS HAVE STRINGS	TENNIS IS A SPORT	SNOW FALLS IN BLIZZARDS	HURRICANES HAVE RAIN
SOCRATES TAUGHT PHILOSOPHY	PLATO WAS A SCHOLAR	SOME FISH ARE DANGEROUS	SHARKS CAN BITE
LEAVES GROW ON BRANCHES	A MAPLE IS A TREE	FAUCETS CAN LEAK	PLUMBERS FIX SINKS
APARTMENTS ARE IN CITIES	A TOWN HAS HOUSES	MATCHES CAN LIGHT FIRES	A CANDLE HAS A FLAME
RABBIT PELTS ARE SOFT	MINKS HAVE FUR	THE PLAINS HAVE GRASS	THE PRARIE IS FLAT
VINES GROW IN JUNGLES	A FOREST HAS TREES	LAWNS CAN BE SOD	GRASS IS GREEN
RADIO STATIONS PLAY MUSIC	A TELEVISION HAS CHANNELS	NURSES WORK IN HOSPITALS	A DOCTOR HAS PATIENTS
AN INFANT OFTEN CRIES	A BABY CAN CRAWL	Weak related fillers	
PENCILS ARE MADE OF WOOD	A PEN HAS INK	A HURRICANE HAS HARD WINDS	SOME STORMS ARE GALES
TRUCKS RIDE ON HIGHWAYS	A CAR HAS TIRES	SOME FISH ARE DANGEROUS	SHARKS HAVE TAILS
A SNAIL CRAWLS	A TURTLE IS SLOW	PLUMBERS FIX SINKS	SEWERS CAN LEAK
SOME BIRDS ARE ACTIVE AT NIGHT	AN OWL CAN HOOT	A CANDLE HAS A FLAME	MATCHES CAN BE LIGHTED
AIR HAS NITROGEN IN IT	OXYGEN IS A GAS	MEADOWS CAN HAVE HILLS	THE PRARIE IS FLAT
THE PACIFIC IS DEEP	THE ATLANTIC IS AN OCEAN	A MOWER CAN CUT A LAWN	GRASS CAN BE SOD
THE NORTH POLE HAS ICE	THE ARCTIC IS COLD	SURGEONS PERFORM OPERATIONS	A DOCTOR HAS PATIENTS
FISH BREATHE THROUGH GILLS	A PERSON HAS LUNGS	Neutral fillers	
MILK COMES IN BOTTLES	JARS ARE GLASS	XXXXXX	A RAISIN IS A GRAPE
RAILROADS CARRY FREIGHT	A TRAIN HAS WHEELS	XXXXX	LADDERS HAVE RUNGS
SAILORS SERVE IN THE NAVY	AN ARMY HAS SOLDIERS	XXXXX	IRON IS HEAVY
FINGERS ARE ON HANDS	A FOOT HAS TOES	XXXXX	A TREE HAS BARK
BURLAP IS A ROUGH FABRIC	SILK IS SMOOTH	XXXXX	HONEY IS STICKY
GREEN PEPPERS ARE VEGETABLES	A TOMATO IS RED	XXXXX	SLEET IS ICE
WEEDS GROW IN DIRT	THE SOIL IS BROWN	XXXXX	A SKYSCRAPER IS TALL
SOME BOOKS ARE FICTION	A NOVEL HAS CHAPTERS	XXXXX	A FLAME IS HOT
PEPPER MAKES PEOPLE SNEEZE	SALT IS A SEASONING	XXXXX	BOOKS HAVE PAGES
ORANGES ARE JUICY	AN APPLE IS A FRUIT	XXXXX	MOTHS HAVE WINGS
A COAT IS WARM IN WINTER	A SWEATER IS WOOL	XXXXX	A TREE HAS SHADE
SHINGLES ARE ASPHALT	A HOUSE HAS A ROOF	XXXXX	A COD IS A FISH
A VIOLIN IS AN INSTRUMENT	A PIANO HAS KEYS	False response fillers	
ACORNS GROW ON OAK TREES	A PINE HAS CONES	BANANAS ARE YELLOW	A PEACH IS A VEGETABLE
TROUSERS ARE PANTS	SHIRTS HAVE SLEEVES	FISH SWIM IN WATER	A PIKE IS A SEAL
A GIRAFFE IS TALL	A ZEBRA HAS STRIPES	TIGERS ARE FIERCE	A LION IS AN INSECT
A GALLERY SELLS PICTURES	A MUSEUM HAS PAINTINGS	A BEE CAN STING	A SPIDER HAS CLAWS
A PILLOW IS SOFT	A BED HAS A MATTRESS	HAMMERS POUND NAILS	DRILLS HAVE CHAIRS
SNOW FALLS IN WINTER	SUMMER IS HOT	REDWOODS ARE TALL	BIRCHES HAVE ELMS
PEOPLE HAVE NOSTRILS	A NOSE CAN SMELL	MOUNTAIN TOPS ARE COLD	A VALLEY HAS A ROOF
Common related fillers		DEER LIVE IN THE WOODS	ANTELOPES ARE BLUE
THIEVES STEAL MONEY	CROOKS ROB BANKS	A GUN FIRES BULLETS	A KNIFE CAN RUN
A COMPOSER WRITES MUSIC	MUSICIANS PLAY SONGS	ROBINS BUILD NESTS	SPARROWS ARE LIZARDS
ARMS BEND AT ELBOWS	LEGS HAVE KNEES	SOME PLANES FLY FAST	A JET CAN SLEEP
A NEEDLE IS SHARP	A PIN HAS A POINT	THE EARTH ROTATES	JUPITER IS A STAR
FARMERS GROW CORN	A FARM HAS FIELDS		

(continued)

Appendix (continued)

Cue	Target	Cue	Target
False response fillers (continued)		False response fillers (continued)	
A CARPET IS SOFT	AN ALLIGATOR IS A BIRD	XXXXXX	BOOKS ARE METAL
PARAGRAPHS HAVE SENTENCES	AN ARM IS A TOE	XXXXXX	A CLOUD IS GREEN
WINDOWS ARE MADE OF GLASS	A TIRE IS SQUARE	XXXXXX	A FENCE HAS WHEELS
ROWBOATS FLOAT ON WATER	FROGS HAVE WINGS	XXXXXX	STATUES HAVE PAGES
A CLOCK HAS HANDS	SUGAR IS SOUR	XXXXXX	A STORE IS A COLOR
A RAZOR CAN CUT	A CACTUS IS RICH	XXXXXX	WINTER IS A MONTH
XXXXXX	A SIDEWALK IS DEEP	XXXXXX	A FLOOR IS TALL
		XXXXXX	ROSES ARE WEEDS
		XXXXXX	A FLAG CAN EAT
		XXXXXX	BUSES HAVE SLEEVES
		XXXXXX	A BRICK IS A PLANT

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